Collatz, but stop at the drop

The game on the grid of $2^j 3^i$

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Abstract

Papers 1 and 2 reduce the question whether every $x > 1$ eventually falls below itself under the reduced Collatz map T to two open statements. This paper reads the admissible words of paper 2 on the grid of the 3-smooth numbers $2^j 3^i$: the odd steps of a word are a staircase on that grid, the constant ρ of the word is the sum of its points, and admissibility is a wall. New here, as far as we know:

1. *The largest ρ.* Of each Order it is a sum of 3-smooth numbers, reached by the first word only, and it shortens the finite reduction of the width identity (both Terras's, 1976); its records are proved to lie exactly at the upper record Orders. With a range check it gives the width identity up to Order 64 497 106.
2. *Powers of* 2 *and* 3. Lean's kernel checks $\min(2^i, 3^j) \le p_2(j)^2\,|2^i - 3^j|$ for every j up to 10^{200}, by a certified walk towards $\log_2 3$ that forms no large power of 3.
3. *The n-th* 3*-smooth number.* A program that forms no 3-smooth number returns the exponents of the n-th one, proved for every n, with $\log_2 3$ replaced by fractions that the program certifies itself.
4. *Three letters.* For the map of OEIS A324245, the stop comes at the drop or earlier, at a merge of paths, and every odd number above 1 stops exactly when every path under T reaches 1: an early stop, equivalent to the Collatz conjecture.

Two hopes fail: the drop of a word does not follow from the drops of its blocks, and a natural strengthening of the width identity, block by block, is false at Order 35. Every result with a Lean name is proved in Lean 4 with Mathlib, without `sorry`; open statements and a published bound enter as named hypotheses, the range check as a hypothesis.

Main results

Words, Orders, class ids, $p_2(n)$, $d_{23}(n) = 2^{p_2(n)} - 3^n$, the upper record Orders $1, 3, 5, 17, 29, \ldots$ and the width identity are those of paper 2; for a word of Order n, $T^{p_2(n)}(x) = (3^n x + \rho)/2^{p_2(n)}$.

Theorem A (The largest ρ; Chapter 3; the bound after Terras)**.** *For every admissible word of Order n,*

$$\rho \le \rho_{\max}(n) = \sum_{m<n} 3^{\,n-1-m}\, 2^{\,p_2(m)-1},$$

with equality only for the first word (rho_le_rhoMax, rho_eq_rhoMax_iff). The width identity holds if and only if, at every Order n, every class id $c > 1$ with $d_{23}(n)\, c \le \rho_{\max}(n)$ drops (classDrops_iff_small_rhoMax); at Order n a range check up to $W(n) = \lfloor \rho_{\max}(n)/d_{23}(n) \rfloor$ gives it (classDropsAt_of_widthMax), and $W(n)$, Terras's bound [23, Thm. 2.10], sets a new record exactly at the upper record Orders (upperRecord_iff_widthBoundMax). With a range check of every $1 < x \le 294\,401\,822\,513\,350$, entered as a hypothesis, the width identity holds at every Order up to $64\,497\,106$ (classDropsAt_le_64497106).

Theorem B (Powers of 2 and 3; Chapter 3)**.** *For $1 \le j \le 10^{200}$ and every i, $\min(2^i, 3^j) \le p_2(j)^2\, |2^i - 3^j|$ (min_le_of_walk_200), checked by Lean's kernel along a walk down the Stern–Brocot tree towards $\log_2 3$ from a bracket $2^a \le 3^K < 2^b$; any walk that reaches Order N gives the same up to N (min_le_of_walks). Without any bound, a congruence sieve proves the first 3000 terms of OEIS A014121 (a014121_iff).*

Theorem C (The n-th 3-smooth number; Chapter 5)**.** *For every $n \ge 1$, a program that computes only floor sums of the numerators and denominators of record fractions returns (j, i) with exactly $n - 1$ of the 3-smooth numbers below $2^j 3^i$; its fraction comes from a Stern–Brocot walk decided in integers, and no convergent of $\log_2 3$ is assumed (latticeOrd_smoothJIFind).*

Theorem D (Three letters; Chapter 7)**.** *For the map of OEIS A324245 on the odd numbers $k \ge 3$, the number M of its steps `U` and `D` up to and including the first value below k is at most the Order of the drop of k under T, with equality exactly when the last step is `D` (le_order_of_isUdsStop, isUdsStop_of_S); and every odd $k \ge 3$ reaches a value below k if and only if every path under T reaches 1 (stops_iff).*

What was known

The formula for T^L, the residue classes of the parity words and the density result are due to Terras [23] and Everett [4]; the admissible words, their chains, the record Orders and the width identity are those of papers 1 and 2 [26, 27] (the width identity is Terras's Conjecture 2.9). Mintz [11] counted the 3-smooth numbers below $2^j 3^i$, the Christoffel words are those of Berstel, Lauve, Reutenauer and Saliola [2], the bound on $2^{p_2(n)} - 3^n$ is Rhin's [21], and Belaga [1] bounds the cycles of $3x + d$. The largest ρ, its unique word and the bound $W(n)$ of Theorem A are Terras's [23, Thm. 2.6, Thm. 2.10], who tabulated where the bound jumps; the estimate $T^L(x) < 3^n x/2^L + n/3$ is Garner's [5]. Shanks [22] and Jackson and Matthews [6] computed the continued fraction of a logarithm; the author's PARI program for the n-th 3-smooth number is in OEIS A003586 [13]. Monks [12] carried T over to the power series over the field with two elements, and the three letters are the map of Vaillant and Delarue [24]. As far as we know, the rest of Theorem A (the records of W proved, and the range), Theorems B to D, the two negative results of Chapter 4 and the formalization (Appendix A) are new.

1 One language for the drop and the 3-smooth numbers

Papers 1 and 2 study the drop: does every $x > 1$ eventually fall below itself under the map T? Paper 2 reduces it to two open statements, the covering and the width identity (paper 2, Chapter 7). This paper collects what lies around them and is proved, and some of what is measured, in one language: the grid of the numbers $2^j 3^i$. *Proved* means proved in Lean where a name is given; *measured* means computed exactly, on the range given.

One grid

The 3-smooth numbers are the points of the grid, sorted by their binary logarithm $j + i \log_2 3$. The admissible words live on the same grid: the n odd steps of a word of Order n are n points, one per row, a staircase that starts at 3^{n-1} and stays to its left (Chapter 2). The exact links between the two:

- $p_2(n)$, the length of a word of Order n, is the binary length of 3^n, and the sum of the $p_2(k)$ is a term of Mintz's [11] count of the 3-smooth numbers below $2^j 3^i$ (card_smoothBelowAt; paper 2, Chapter 4).
- The record Orders of the block factor $f(n) = 3^n / 2^{p_2(n)}$ decide the blocks of the chain of a word (paper 2), and also the step from a 3-smooth number to the next (smooth_next, smooth_succ; Chapter 2).
- ρ, the constant of $T^L(x) = (3^n x + \rho)/2^L$, is the sum of the staircase's n points, and $d_{23}(n) = 2^{p_2(n)} - 3^n$ is a difference of two 3-smooth numbers. The width identity says that the class id times this difference exceeds the sum of the staircase (ClassDrops, widthIdentity_iff_classDrops).

Whether this language moves either open statement is not known; it is a common language, not yet a tool.

The Lean files, the scripts and this text were written with the assistance of Claude (Anthropic) (Appendix A); the author checked the statements and takes responsibility for them.

The chapters

Chapter 2 draws the words as staircases and bounds ρ. Chapter 3 takes the largest ρ of each Order as the bound of the finite reduction of the width identity, with its records. Chapter 4 asks whether the congruences that fix a class id carry the drop, and finds that they restate it. Chapter 5 finds the n-th 3-smooth number without listing the others. Chapter 6 computes the smallest class id of each Order, and Chapter 7 plays the game with three letters. Chapter 8 is an open statement on neighbouring 3-smooth numbers, with its reductions, and Chapter 9 says what is left out.

Key terms introduced in Chapter 1	
proved	proved in Lean where a Lean name is given, otherwise on paper where the argument is given.
measured	computed exactly, in integers and rational numbers (no floating point), on the range given; not proved.

2 Words as staircases

A word of Order n has n odd steps; let the o-th stand at position k_o, counted from 0. After the whole word, $T^L(x) = (3^n x + \rho)/2^L$ with

$$\rho = \sum_{o=1}^{n} 3^{\,n-o}\, 2^{\,k_o}$$

(paper 2, Chapter 3). Each term is a 3-smooth number $2^j 3^i$, with $i = n - o$ and $j = k_o$. From one odd step to the next the exponent of 3 falls by one and the exponent of 2 rises: the n terms are a *staircase*, one point in each row $i = n-1, \ldots, 0$ of the grid of the 3-smooth numbers. As the exponents of 2 rise while those of 3 fall, no term divides another, and with $k_1 = 0$ the sum $\rho = 3^{n-1} + 3^{n-2}2^{k_2} + \cdots + 2^{k_n}$, $0 < k_2 < \cdots < k_n$, is what Blecksmith, McCallum and Selfridge [3] call a special 3-smooth representation of level $n-1$, one in which every power 3^i with $i < n$ appears; they note its link with the $3x+1$ problem.

The slanted grid

Put $2^j 3^i$ at the horizontal position $j + i \log_2 3$, its binary logarithm, in the row i: the *slanted grid*. Then the points of the grid stand from left to right in the order of their values, and a vertical line through a point separates the smaller 3-smooth numbers from the larger ones (the logarithm only places the points; every comparison below is one of integers). In this picture admissibility is a *wall*. In an admissible word the odd step that follows $m \geq 1$ odd steps stands at a position k with $k + 1 \leq p_2(m)$, that is $2^k < 3^m$, so its point $2^k\, 3^{n-1-m}$ is below 3^{n-1}; the first odd step stands at 0, and its point is 3^{n-1} itself. Together with the length $p_2(n)$ this is also enough: a word is admissible exactly when its staircase starts at 3^{n-1}, every later point lies left of 3^{n-1}, and its length is $p_2(n)$ (admissible_iff_steps, admissible_step_le). Figure 1 shows the first word of Order 5.

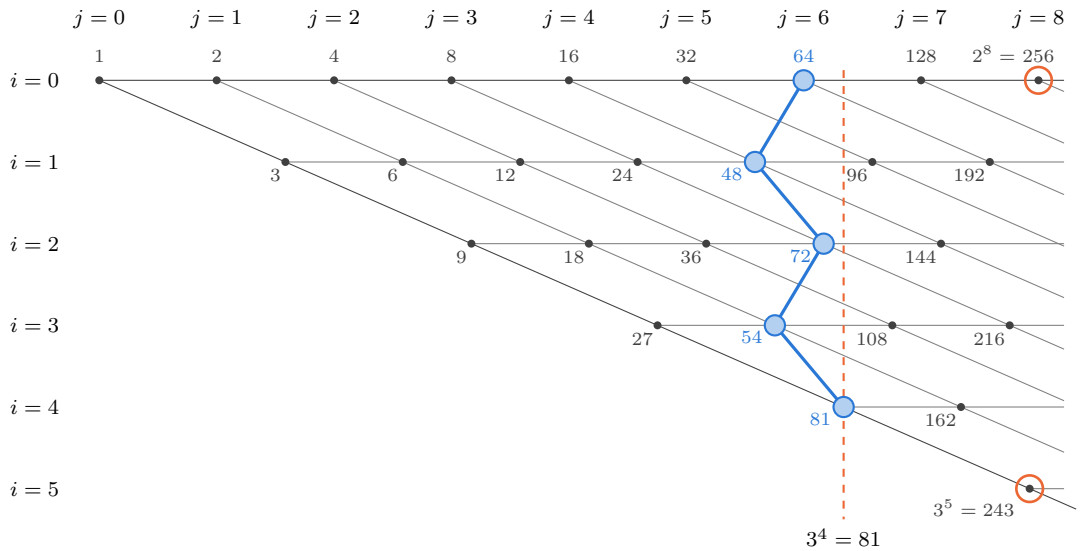


Figure 1: The first word 11011010 of Order 5 as a staircase on the slanted grid: its odd steps at positions $0, 1, 3, 4, 6$ are the points 81, 54, 72, 48, 64, and $\rho = 319$. Each grid point $2^j 3^i$ carries its value. Dashed: the wall at 3^4. Ringed: 3^5 and $2^{p_2(5)} = 2^8$, whose difference 13 is d_{23}.

The bounds on rho

Every point of the staircase is at most 3^{n-1}, and only the first reaches it, so $\rho < n\,3^{n-1}$ from Order 2 on (rho_lt_of_admissible); at Order 1 the word 10 has $\rho = 1$; Chapter 3 finds the largest ρ, where every point stands as far right as the wall allows. From below, $k_o \geq o - 1$ in every word, so each point is at least $3^{n-o}\,2^{o-1}$ and $\rho \geq \sum_o 3^{n-o}\,2^{o-1} = 3^n - 2^n$, with equality exactly for the words that start with all their odd steps (three_pow_le_rho_add, rho_add_eq_three_pow_iff). The staircase of such a word is the diagonal from 3^{n-1} down to 2^{n-1}.

The two ends

The cycle point $x^* = \rho/d_{23}$ has $d_{23}(n) = 2^{p_2(n)} - 3^n$, a difference of two 3-smooth numbers: 3^n at the start of row n, and $2^{p_2(n)}$, the first power of 2 beyond it. A 3-smooth number $2^j 3^i$ lies strictly between them exactly when $1 \leq i < n$, $j = p_2(n) - p_2(i)$ and the block factor of i is above that of n (smooth_between_iff). So none lies between them exactly when n is an upper record Order (upperRecord_iff_none_between): in Figure 1, nothing lies between 243 and 256; for $n = 2$, the number $12 = 2^2 \cdot 3$ lies between 9 and 16.

Walking up through the 3-smooth numbers from 3^n to $2^{p_2(n)}$, each step is one of the two of paper 2, Chapter 4: from $2^j 3^i$ the exponent of 3 falls by the last upper record Order $b \leq i$ ($b = 0$ is the doubling), or rises by the last lower record Order a with $p_2(a) \leq j+1$ (smooth_succ). After a power of 3 it is always the step down by b (smooth_succ_of_pow_three); at an upper record Order n that one step, with $b = n$, goes from 3^n straight to $2^{p_2(n)}$.

Over the power series in q

K. G. Monks [12] carries T over to the power series in q over the field with two elements: for A and B with constant term 1, $T_{A,B}(x) = x/q$ when q divides x, and $(Ax + B)/q$ otherwise; each $T_{A,B}$ is conjugate to T by a map that keeps the parities. The staircase goes with it. If x has n odd steps among its first L steps, the o-th at position k_o, then

$$q^L\,T_{A,B}^L(x) = A^n x + B\rho_A, \qquad \rho_A = \sum_{o=1}^{n} A^{n-o}\,q^{k_o},$$

the formula for ρ with 2 replaced by q and 3 by A, and no signs, as $1 + 1 = 0$ (X_pow_mul_monksT_iterate). For $A = 1$ the map is the shift σ (which deletes the first coefficient) conjugated by B: $T_{1,B}(By) = B\,\sigma(y)$ (monksT_one_mul). Monks lists the series that correspond to $x = 1, \ldots, 20$ for $A = 1 + q$, $B = 1$ and for $A = 1$, $B = 1 + q^2$, and asks for a closed formula. For a start x that reaches 1 after L steps, 1 corresponds to the point $B/(A+q^2)$ of the 2-cycle of $T_{A,B}$, so its series is $(q^L B/(A + q^2) + B\rho_A)/A^n$; for $A = 1$, $B = 1 + q^2$ this is the polynomial $q^L + (1 + q^2)\sum_o q^{k_o}$, of degree L (checked for all 20 rows of both tables). With the coefficients as binary digits, the second table is OEIS A389685 [19], the value at $q = 2$ of this polynomial (checked for the first 10^4 terms). The closed formula is the orbit itself.

Key terms introduced in Chapter 2	
staircase	the n terms $3^{n-o}\,2^{k_o}$ of ρ of a word of Order n, one point in each row of the grid of the 3-smooth numbers.
slanted grid	$2^j 3^i$ drawn at $j + i\log_2 3$ in row i: the points stand in the order of their values.
wall	the vertical line through 3^{n-1}; a word is admissible when its staircase starts at 3^{n-1}, every later point lies left of it, and its length is $p_2(n)$.

3 The largest rho and the finite reduction

Paper 2 bounds ρ by $n\,3^{n-1}$ for every admissible word of Order n, and so the width identity needs checking only at class ids up to $\tau(n) = n\,3^{n-1}/d_{23}(n)$ (paper 2, Chapter 7). This chapter replaces $n\,3^{n-1}$ by the largest ρ that an admissible word of Order n really has, a sum of 3-smooth numbers, and shows that the finite reduction keeps its shape: its records are again exactly the upper record Orders. For cycles of $3x + d$ with k odd steps, Belaga bounds the least member m by $2^\ell \le (3 + d/m)^k$, with $\ell \ge p_2(k)$ for the bound in k alone, by averaging over the rotations of the word, and so, with linear forms in logarithms, by $d\,k^C$ [1] (a short proof by another route: [7]). At $d = 1$, Bernoulli's inequality gives $2^\ell = 3^k + d_{23}(k) \le (3 + 1/\tau(k))^k$, so $\tau(k)$ obeys the same bound.

The largest rho

In $\rho = \sum 3^{n-o}\,2^k$ the o-th odd step stands at position k. In an admissible word the odd step that follows m odd steps stands at a position k with $k + 1 \le p_2(m)$ (admissible_step_le): before it the level of paper 2, the running product of $3/2$ per 1 and $1/2$ per 0, is at least 1. Each term grows with its position, so ρ is largest when every odd step stands as late as it may, at $k = p_2(o-1) - 1$. That gives

$$\rho_{\max}(n) = \sum_{m<n} 3^{\,n-1-m}\,2^{\,p_2(m)-1}, \qquad \rho_{\max}(n+1) = 3\,\rho_{\max}(n) + 2^{\,p_2(n)-1},$$

and $\rho \le \rho_{\max}(n)$ for every admissible word of Order n (rho_le_rhoMax, rhoMax_eq_sum). Every term is a 3-smooth number: in the grid of Chapter 2, each odd step stands as close to the line as admissibility allows. The one admissible word with every odd step at its latest position is the first word of Order n (rank 1), and it is the only word that reaches the bound (rho_eq_rhoMax_iff, rho_lt_rho_firstWord). This is Terras's Mini-Max theorem [23, Thm. 2.6], with the first word as his sequence ν of Theorem 2.4 there; here it is proved in Lean. For $n = 3$ the first word is 1 1 0 1 0, with odd steps at 0, 1 and 3, and $\rho_{\max}(3) = 9 + 6 + 8 = 23$, against $n\,3^{n-1} = 27$.

At an upper record Order the first word is an upper Christoffel word [2, Definition 1.1], at a lower one such a word followed by 0: with $P = p_2(n)$ at an upper record Order n, and $P = p_2(n)-1$ at a lower one, its o-th odd step stands at $\lfloor (o-1)P/n \rfloor$, by $\lfloor iP/Q \rfloor + 1 = p_2(i)$ for a record fraction P/Q (Chapter 5); with 0 as x and 1 as y these P letters are the upper Christoffel word of slope $n/(P-n)$ (take_firstWord_recordFrac; n and P are coprime, RecordFrac.coprime: a common factor would give an earlier record), and at a lower record Order a 0 follows. Below Order 200 the first word has this form at no other Orders than these and 4, 6, 24, 36, 48, 106, 159, where $(p_2(n)-1)/n$ is the record fraction of a lower record Order, not in lowest terms, and the first $p_2(n) - 1$ letters a power of that Order's Christoffel word (measured).

What the bound $n\,3^{n-1}$ gives away, its *slack*, is half a sum of the lower deficits $2\cdot 3^m - 2^{p_2(m)}$:

$$2\left(n\,3^{n-1} - \rho_{\max}(n)\right) = \sum_{m<n} 3^{\,n-1-m}\left(2\cdot 3^m - 2^{p_2(m)}\right)$$

(rhoMax_slack; for $n = 3$: $0 + 6 + 2 = 8$). Two of its terms compare like $(2\cdot 3^m - 2^{p_2(m)})\,3^k$ and $(2\cdot 3^k - 2^{p_2(k)})\,3^m$ (slackTerm_lt_iff), and the term of m is below every earlier one from $k = 1$ on exactly when m is a lower record Order (lowerRecord_iff_slackTerm). So the upper record Orders are where $d_{23}(n)/3^n$ is smallest so far, the lower ones where $(2\cdot 3^m - 2^{p_2(m)})/3^m$ is.

The finite reduction with the largest rho

A class id c of a word of Order n drops at its end when $\rho < d_{23}\,c$, so only the class ids with $d_{23}\,c \le \rho_{\max}(n)$ can fail:

$$\text{the width identity} \iff \text{every class id } c > 1 \text{ with } d_{23}\,c \le \rho_{\max}(n) \text{ drops}$$

(classDrops_iff_small_rhoMax). Per Order, the class ids up to

$$W(n) = \left\lfloor \frac{\rho_{\max}(n)}{d_{23}(n)} \right\rfloor \le \lfloor \tau(n) \rfloor$$

suffice, and a range check of every x with $1 < x \le X$ gives the width identity at every Order with $W(n) \le X$ (classDropsAt_of_widthMax, classDropsAt_of_verifiedMax, widthBoundMax_le_widthBound). The bound $\rho_{\max}(n)/d_{23}(n)$ is Terras's [23, Cor. 2.8, Thm. 2.10], in his notation $\omega/(2-\mu)$ (equal as fractions at every upper record Order from 3 to 1636, computed); his Table C lists its jumps up to length 2593, at the lengths $p_2(n)$ of the upper record Orders, each as $W(n)+1$ (72 059 for $n = 306$).

The records

The ratio $\rho_{\max}(n)/d_{23}(n)$, and its floor $W(n)$, are above their values at every earlier Order exactly at the upper record Orders (upperRecord_iff_rmax, upperRecord_iff_widthBoundMax), as for τ: Terras's Table C shows it up to length 2593, and here it is proved. Both factors of

$$\frac{\rho_{\max}(n)}{d_{23}(n)} = \frac{\rho_{\max}(n)/3^n}{1/f(n) - 1}$$

work the same way: $\rho_{\max}(n)/3^n$ grows with n (three_pow_mul_rhoMax_le), and $1/f(n) - 1$ falls as the block factor $f(n)$ of paper 2, Chapter 4, rises. So a later Order with a larger block factor has a larger ratio (rmax_lt_of_lt_of_blockFactor_lt; two Orders never have the same block factor, as $3^n/2^{p_2(n)}$ fixes n), and an upper record Order has a ratio above every earlier one.

For the other direction, let n be no upper record Order, a the last upper record Order below it and $b = n - a$. Then p_2 is additive at every split $a + j$ with $1 \le j \le b$ (p2_of_add_of_upperRecord), and

$$\rho_{\max}(a+b) = 3^b\,\rho_{\max}(a) + 2^{p_2(a)}\,\rho_{\max}(b) - 3^{\,b-1}\,2^{\,p_2(a)-1}, \qquad d_{23}(a+b) = 3^b\,d_{23}(a) + 2^{p_2(a)}\,d_{23}(b)$$

(rhoMax_add). Without the last term the ratio at $a + b$ would be a mediant of the ratios at a and b, with the weights 3^b and $2^{p_2(a)}$; the term takes it below. With the ratio at b at most that at a (by induction), the ratio at n is below that at a (rmax_add_lt). That the floor W has the same records takes one more step: for two Orders $c < n$ with $f(3) \le f(c) \le f(n)$, the ratio at n is at least 1 above that at c (rmax_add_one_le).

Table 1 puts $W(n)$ beside $\lfloor \tau(n) \rfloor$ at 3 and the upper record Orders of paper 2's Table 6. With the last upper record Order a up to n, W of every Order up to n is at most $W(a)$ (widthBoundMax_le_of_upperRecord), so a range check up to $W(a)$ gives the width identity at every Order up to n (classDropsAt_of_verifiedMax_record). The range check of paper 2 could have stopped at $W(190\,537)$ instead of $\lfloor \tau(190\,537) \rfloor$; one up to $\lfloor \tau(10\,781\,274) \rfloor = 294\,401\,822\,513\,350$ gives it up to Order 64 497 106 (classDropsAt_le_64497106).

n	$p_2(n)$	$\lfloor \tau(n) \rfloor$	$W(n)$
3	5	5	4
17	27	144	108
41	65	1 185	867
306	485	99 729	72 058
15 601	24 727	285 814 986	206 178 386
47 468	75 235	1 447 674 321	1 044 289 613
79 335	125 743	7 216 089 270	5 205 340 379
190 537	301 994	984 572 779 224	710 220 447 737

Table 1: The bound of paper 2 and the bound with the largest ρ, at some upper record Orders (computed exactly).

The bound from Rhin, checked to Order 10^{60}

Paper 2 cites Rhin's [21] lower bound for $2^{p_2(n)} - 3^n$ as a hypothesis (RhinBound) and notes that Lean's kernel checks $3^n \le p_2(n)^2\,(2^{p_2(n)} - 3^n)$ for every $n \le 10^{60}$ (rhinBoundUpTo_one_two). This is a finite check, and no proof uses it; what is of interest is how it avoids the powers 3^n with n near 10^{60}. In product form, with $Q = p_2(n)$, the bound reads $3^n\,(1 + Q^2) \le Q^2\,2^Q$. Between upper record Orders the block factor stays at most its value at the last record a, and p_2 does not fall; so the bound at a gives it at every Order up to the next record (rhinAt_of_record, rhinBoundUpTo_of_records). The upper record Orders are the upper denominators of the walk down the Stern–Brocot tree towards $\log_2 3$ (paper 2, Figure 7), and at each such fraction p/n with $p = p_2(n)$ the check at the fraction computes no power 3^n with $n > 100$: for $n \le 100$ it is exact, and otherwise it comes from a bracket $3^K < 2^b$ with $K = 2^{400}$. Raised to the power K, the bound follows from $b\,n + \lfloor K/\lfloor p^2/2 \rfloor \rfloor + 1 \le p\,K$, a comparison of numbers of at most 181 digits, by $(1 + p^2)^K \le 2^{\lfloor K/\lfloor p^2/2 \rfloor \rfloor + 1}\,p^{2K}$ (rhinOK_spec, succ_pow_le). The walk itself is certified with the same bracket: 649 steps until the sum of its two denominators passes 10^{60}, with 320 upper record Orders among them (rhinWalk_ok, computed by the kernel; the two counts measured).

The lower side, and the differences of powers of 2 and 3

The Rhin check bounds the gap $2^{p_2(n)} - 3^n$ above 3^n. The same walk, at its lower fractions, bounds the gap $3^n - 2^{p_2(n)-1}$ below: with $Q = p_2(n)$,

$$2^{\,Q-1}\,(1 + Q^2) \le Q^2\,3^n \qquad (1 \le n \le 10^{60})$$

(lowAt_upTo). Between lower record Orders it carries over as the upper side does between upper ones (lowAt_of_records), and the walk checks the lower fraction of every node with the same bracket (lowWalk_ok, computed by the kernel). With both sides, no power of 2 comes close to a power of 3:

$$\min(2^i, 3^j) \le p_2(j)^2\,|2^i - 3^j| \qquad (1 \le j \le 10^{60})$$

(min_le_of_walk), so a number $v = |2^i - 3^j|$ with $j \le 10^{60}$ has $i, j < B$ for every $B \ge 3$ with $(4B^2 + 1)\,v < 2^B$ (lt_of_walk).

Nothing in these proofs depends on the bound 10^{60} but the walk itself: any walk at $K = 2^m$ that checks every node until its denominators pass N gives both sides up to Order N (rhinBoundUpTo_of_walk, lowAt_of_walk, min_le_of_walks). The bracket has to grow with N: a walk to 10^{100} needs $m = 670$ and 1063 steps, one to 10^{200} needs $m = 1330$ and 5839 steps (computed in integers, with the bracket from $2m + 3$ bits as in log2Lo): in each case $K = 2^m$ is just above N^2 (m tried in steps of 10). The kernel has run both walks to 10^{200}, each in four stretches (rhinBoundUpTo_one_two_200, lowAt_upTo_200), from a bracket whose 1330 squarings it checks in fourteen chunks, each from the state before it written out (farLo200_eq, farHi200_eq); so $\min(2^i, 3^j) \le p_2(j)^2\,|2^i - 3^j|$ holds for $1 \le j \le 10^{200}$ (min_le_of_walk_200).

These differences are the terms of OEIS A014121 [14] ($i, j \geq 0$) and A363999 [18] ($i, j \geq 1$). Beyond 10^{200} the bound needs Rhin's, which enters in integers, as the hypothesis RhinInt: when 2^i and 3^n are within a factor 2 ($i \geq 2$), the smaller is at most $i^{14}\,|2^i - 3^n|$. It follows from Rhin's bound by $\min(a, b)\,|\log(a/b)| \leq |a - b|$, and assumes less. Under it a finite search decides every v (isPowDiff_iff_searchB). Without any bound, a congruence sieve proves the first 3000 terms of A014121, up to $N = 328\,256\,967\,394\,000\,206\,715$, and the 2891 terms of A363999 up to N (a014121_iff, a363999_iff): modulo $M = 2^8\,3^3\,p_1 \cdots p_{13}$, with the 13 primes $5, 7, 13, \ldots, 1153$ at which the orders of 2 and 3 divide 288 and 1728, the residue of $2^i - 3^j$ for $i \geq 8$, $j \geq 3$ is one of $288 \cdot 1728$, and each of those within N of 0 is the residue of a term; as $2N < M$, the residue gives the number. The cases $i < 8$ or $j < 3$ are checked directly.

Key terms introduced in Chapter 3	
$\boldsymbol{\rho_{\max}(n)}$	the largest ρ of an admissible word of Order n, reached by the first word only.
slack	$n\,3^{n-1} - \rho_{\max}(n)$, half a sum of lower deficits.
$\boldsymbol{W(n)}$	$\lfloor \rho_{\max}(n)/d_{23}(n) \rfloor$: the class ids of Order n up to it decide the width identity there.

4 The drop as a congruence

A word w of Order n and length $L = p_2(n)$ has its class id c, its ρ and its $d_{23} = 2^L - 3^n$, and a member x of its class drops at the end of the word exactly when x lies above the cycle point $x^* = \rho/d_{23}$ (paper 2, Chapter 3). The class id is fixed by congruences, one for each piece of the word. This chapter asks whether those congruences carry the drop, and finds that they restate it.

Pieces of a word

For any two words u and v (not necessarily admissible), with n_u odd steps among L_u steps, and so on,

$$\rho(uv) = 3^{n_v}\rho(u) + 2^{L_u}\rho(v)$$

(rho_append), and likewise $2^L - 3^n$ of uv is $3^{n_v}(2^{L_u} - 3^{n_u}) + 2^{L_u}(2^{L_v} - 3^{n_v})$. A number with the steps uv has the steps u, and after them the steps v: so the class id c of uv is congruent to the class id of u modulo 2^{L_u}, and $T^{L_u}(c)$ is congruent to the class id c_v of v modulo 2^{L_v}.

Not block by block

An admissible word splits at its record lows into the blocks of its chain, $w = \boxed{1}B_1 \cdots B_k$ (paper 2, Chapter 4), and each block is an admissible word of lower Order. In one sentence: after the first $\boxed{1}$, each block is a small, local drop that ends below where it began (by a factor between 1/2 and 1, one_lt_deficitRatio, deficitRatio_le_two); the blocks use up the 3/2 of that first $\boxed{1}$, and the word drops when that budget is spent (validOrders_iff_budget; at most one $\boxed{0}$ follows the last block). One could hope that the drop of w follows from the drops of its blocks: that c, carried through the word, passes above the cycle point of each block in turn. It does not, and the cycle point itself shows why. For $w = \boxed{1}\boxed{1}\boxed{0}\boxed{0}$ ($c = 3$, $x^* = 5/7$) the blocks are $\boxed{1}\boxed{0}$ (cycle point 1) and $\boxed{0}$ (cycle point 0). Started at 5/7, the word reaches $11/7 > 1$ before its first block and $10/7 > 0$ before its second: 5/7 passes every block's test, and yet it does not drop, it comes back. So no argument block by block can give the drop.

The bound of a split

What the congruences do give is a bound, the *bound of a split*. Split $w = uv$. As $T^{L_u}(c)$ is a positive number in the class of v, it is at least c_v, the least one; and $T^{L_u}(c) = (3^{n_u}c + \rho(u))/2^{L_u}$, so

$$c \;\geq\; X_u = \frac{2^{L_u}c_v - \rho(u)}{3^{n_u}}.$$

When $X_u > x^*$, the drop at c follows. The bound has a plain meaning. The steps u send the cycle point of uv to the cycle point of the rotation vu, and so

$$X_u > x^*(uv) \iff c_v > x^*(vu)$$

(split_bound_iff, from the identity split_identity in integers). And for the split after the leading $\boxed{1}$, $w = \boxed{1}v$: when $3c+1 \leq 2^L$, the number $T(c) = (3c+1)/2$ is at most 2^{L-1} and has the steps v, so $T(c) = c_v$ (classId_tail_eq_tStep), and the bound of this split is the drop at c itself (split_one_iff). The class ids that could break the width identity are at most $\lfloor n3^{n-1}/d_{23} \rfloor$ (paper 2), and these satisfy $3c + 1 \leq 2^L$ at every Order up to 10^{60} (three_mul_widthBound_add_one_le_upTo), and at every Order under Rhin's bound (three_mul_widthBound_add_one_le_of_rhin): with $3^n \leq L^C\, d_{23}$ they are at most nL^C, and $3L^{C+1} < 2^L$ once L is large, the Orders below that checked by the kernel. There the congruence bound is the width identity in other words.

Measured, on every admissible word of Orders 2 to 12: whenever some split at the end of a block gives $X_u > x^*$, the split after the leading $\boxed{1}$ does. On Orders 2 to 18 only four words have no split that gives the drop: $\boxed{1}\boxed{1}\boxed{0}\boxed{1}\boxed{0}$ and $\boxed{1}\boxed{1}\boxed{1}\boxed{0}\boxed{0}$ (Order 3), $\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{0}\boxed{1}\boxed{0}\boxed{0}$ and $\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{1}\boxed{0}\boxed{0}\boxed{0}$ (Order 5). All four have $3c + 1 > 2^L$, at Orders where d_{23} is small (5 and 13).

The cycle points of the blocks

Measured on Orders 1 to 18: the class id of w lies above the cycle point of every block of its chain. This does not follow from the drop of w: on Orders 2 to 13 the cycle point of w is often below that of a block (for every word of Orders 7 and 12). Nor from the drops of the blocks, as c is not always above the class ids of its blocks. It does follow from a statement per block, measured on Orders 2 to 31 (from Order 14 on by a walk over the prefixes): the class id of the prefix $\boxed{1}\,B_1 \cdots B_i$ lies above the cycle point of B_i, as w and that prefix have the same residue modulo 2^{L_P}, L_P the length of the prefix. But that statement holds the drop of B_i itself whenever T^{L_u} of the prefix's class id is the class id of B_i. So it is no easier than the width identity.

Neither statement is true: both fail at Order 35, past those measurements. The word w of 31 is $\boxed{1}\,B\,\boxed{0}$, with B the word of 47, of Order 34 and length 54, the first block of the chain of w. Here $c = 31$, $d(B) = 1\,337\,216\,809\,815\,415$ and $\rho(B) = 44\,834\,791\,551\,842\,521$, so $31\,d(B) < \rho(B)$: the class id lies below the cycle point of its block (not_prefixBlockBound_47, not_lt_classId_31). The drop of w is not touched: 31 drops to 23. Where $T(c) = c_B$, the statement asks c_B to exceed about 3/2 times the cycle point of B, more than the width identity, and at the word of 47 that quotient is $62\,849\,190\,061\,324\,505/44\,834\,791\,551\,842\,521 < 3/2$, the smallest over all words with class id below $2 \cdot 10^{10}$. A scan of the starts below 10^{11} finds no other failure, at any block. So what follows measures how far the statement holds, not a way to the width identity.

The failure has a plain form in a start and its drop. Let $x = c_B$, of Order n, and $y = T^L(x)$ its drop, $L = p_2(n)$; then $x - y = d_{23}(x - x^*)/2^L$, with x^* the cycle point of B. Where $T(c) = x$, so $c = (2x-1)/3$ and $x - c = (x+1)/3$, the statement fails exactly when

$$3 \cdot 2^L\,(x - y) < d_{23}\,(x + 1).$$

For 47 this reads $3 \cdot 2^{54} < 48\,d(B)$, that is $16\,d(B) > 2^{54}$. Below $3 \cdot 10^6$ no other $x \equiv 3 \pmod 4$ meets it, even without $x \equiv 2 \pmod 3$. The gap ratio $(x-y)/x$ is $(1 - 3^n/2^L)(1 - x^*/x)$ (gap_ratio_eq), and the failure needs it small while $d_{23}/2^L$ is not. Its record lows over $x \geq 2$ ($x = 1$ is left out: the path stops at the bottom 1, so drop(1) = 1 and the gap ratio of 1 is 0, which no $x \geq 2$ reaches), as continued fractions, are OEIS A398772 [20], at the record starts $2, 3, 5, 11, 39, 47, 4591, \ldots$; each record start is the least start of its word, as of two starts of one word the smaller has the smaller gap ratio (gap_identity, gap_mul_lt_of_lt). This gives a sieve for the search: after the starts up to 2^s, a start $x > 2^s$ whose first $j \leq s$ steps form an admissible word (fixed by x mod 2^s, admissible_tBits_of_modEq) first drops at step j, and the class id of that word, or 5 when it is 1, drops there too, is at most 2^s and has a smaller gap ratio (at Order 0, the word $\boxed{0}$ of the even starts, both have 1/2); so x is no record, and only the residues modulo 2^s with no admissible prefix are scanned further. This needs the drop of the class id at the Order of the word, at most 10 for $s \leq 16$, where it holds unconditionally (sieve_skip_lt, sieve_skip_16_lt; sieve_skip_16 with the tie). An even x drops at once, to $x/2$, and an $x = 4i + 1$ after two steps, to $3i + 1$ ($i = 0$ is drop(1) = 1); up to Order 64 497 106, where every class id above 1 drops given the range check of Chapter 3, each record start $x \equiv 3 \pmod 4$ is the class id of its word (eq_classId_of_gapRecord_mod4_le_64497106). Below 2^{46}, 47 is the only one among them that breaks the statement (measured). There are 13 record starts below 2^{46}, and from 11 on, every one but 47 has an upper record Order, one where $d_{23}/2^L$ reaches a new low $(3, 5, 41, 94, 147, 200, 200, 253, 306)$. Up to 971 those Orders are $1, 3, 5, 17, 29, 41, 94, 147, 200, 253, 306, 971$ (upperRecord_iff): from 41 to 306 in steps of 53, as $2^{84} < 3^{53} < 2^{85}$ keeps $p_2(n+53) = p_2(n)+84$ for $n < 306$, and $p_2(359) = p_2(306)+85$ ends them (pT_add_53, pT_359). Below 2^{46}, at each of these Orders $d_{23}/2^L$ falls so far that every start of the Order beats the previous record, so the least one, A122442, is a record; a second start of the same Order can follow when x^*/x is larger, as 604 499 583 after 371 871 359 at Order 200; only 47, at Order 34, is a record by lying close to its cycle point (measured).

By the index i of the block, the statement gets easier fast. On Orders 2 to 21 the smallest ratio of the class id of $\boxed{1}\,B_1 \cdots B_i$ to the cycle point of B_i is

$$\tfrac{35}{23},\quad \tfrac{2015}{227},\quad \tfrac{19643}{259},\quad \tfrac{13041}{5},\quad \tfrac{1700937}{5}$$

for $i = 1, \ldots, 5$; the first three are reached at Orders 4, 7 and 12, and stay the smallest at every higher Order measured there; further out they do fall, below 6 for $i = 2$ at the word of 27 (Order 37) and below 30 for $i = 3$ at the word of 155 (measured). The cycle point of a block is at most $\rho_{\max}/d_{23}$ of its Order, which under Rhin's bound grows like a power of the Order (paper 2, Chapter 7), while the class id of a longer prefix is fixed by more bits. So the first block is where the statement is tight, and there it contains the width identity: every admissible word B is the first block of the words that begin with $\boxed{1}\,B$, and when c, the class id of $\boxed{1}\,B$, has $3c+1 \le 2^{1+L_B}$, the class id of B is $T(c) > c$ (classId_tail_eq_tStep); so c above the cycle point of B gives the drop of B. (On Order 14, $3c+1 \le 2^{1+L_B}$ holds for 5927 of the 17 637 words, measured.)

The drop amount over the blocks

What does go block by block is the amount of the drop, in integers. Let c be the class id of $w = \boxed{1}\,B_1 \cdots B_k$, of Order n, and $k(c) = c - T^L(c)$ its drop amount; the width identity is $k(c) \ge 1$. Let v_i be the value at the start of B_i, so $v_1 = T(c) = c + (c+1)/2$ and $v_{k+1} = T^L(c)$. As v_i has the steps B_i, $v_i = c_i + q_i\, 2^{L_i}$ with c_i the class id of B_i and $q_i \ge 0$, and $v_i - v_{i+1} = (v_i d_i - \rho_i)/2^{L_i}$, with d_i and ρ_i those of B_i (for a block $\boxed{0}$, the class of the even numbers, take $c_i = 0$, so $q_i = v_i/2$). So

$$k(c) \;=\; \sum_{i=1}^{k} \big(k(c_i) + q_i\, d_i\big) \;-\; \frac{c+1}{2},$$

where $k(c_i) = (c_i d_i - \rho_i)/2^{L_i}$ is the drop amount of the block's own class id, and $(c+1)/2$ the rise of the leading $\boxed{1}$. For the word of 27: $T(27) = 41$ starts the block $\boxed{10}$ ($c_1 = 1$, $q_1 = 10$, $d_1 = 1$), and 31 starts the block of Order 35, the word of 31 itself ($q_2 = 0$); so $k(27) = 10 + k(31) - 14 = 4$. Likewise $k(31) = k(47) + 23 - 16 = 8$ and $k(47) = k(71) + 15 - 24 = 1$: the tight class ids hand their margins on along the path of 27, read backwards.

This gives an induction on the Order that settles part of the words. Suppose $c_B d_B \ge \rho_B$ for every admissible word B of Order below n, so $k(c_i) \ge 0$. Then $k(c) \ge \sum_i q_i d_i - (c+1)/2$, and $c\, d_{23} \ge \rho$ follows for every word of Order n with $\sum_i q_i d_i \ge (c+1)/2$; equality would make c a point of a cycle. In particular this holds for every word with a block $\boxed{0}$: its start value v is even and above c (every proper prefix of an admissible word has $3^m/2^j \ge 1$), and that block alone gives $v/2 \ge (c+1)/2$. The two blocks of $d = 1$, $\boxed{0}$ ($2^1 - 3^0$) and $\boxed{10}$ ($2^2 - 3^1$), pay their whole q; by Levi ben Gershon [9] they are the only ones, as $3^b + 1 = 2^a$ only for $(a, b) = (1, 0)$ and $(2, 1)$, and $2^a + 1 = 3^b$ only for $(1, 1)$ and $(3, 2)$ (three_pow_add_one_eq_two_pow, two_pow_add_one_eq_three_pow: powers of 3 are 1 or 3 modulo 8, and $3^{2k} - 1 = (3^k - 1)(3^k + 1)$). They carry most of the payments: 58% of them over the class ids below $2 \cdot 10^6$, measured.

Measured on the 56 507 class ids below $2 \cdot 10^6$: 32 698 are settled this way, 16 432 of them by a block $\boxed{0}$. Taking the blocks with $q_i = 0$ apart in turn, with the same bound inside, settles 10 970 more at the second level, 5283 at the third, and so on; but no fixed depth is enough: 1 441 407 (Order 94) needs 28 levels, and the deficit handed down does not shrink level by level but rides the climb of the path to its top. The reason is exact. For the 8847 words of the form $w = \boxed{1}\,B$, with B the whole word of $v = T(c)$ and no block that pays (39, 159, 255, 447, ...), $k(c) = k(v) - (c+1)/2$: what v must deliver is $k(v) \ge (v+1)/3 + 1$, that v drops below $(2v-1)/3 = c$, and that is the width identity of c itself. So the induction over the blocks closes only with the statement it is meant to prove as its hypothesis.

With the zeros kept

Halving an even number only sheds a trailing 0-bit. Keeping them, the *zeros kept*, makes the odd step $Z \mapsto 3Z + 2^{\nu(Z)}$, where $2^{\nu(Z)}$ is the lowest set bit of Z: for $Z = 2^k v$ with v odd it gives $2^k(3v+1)$. From $Z_0 = c$, after n odd steps

$$Z_n = 3^n c + \rho,$$

and the points $3^{n-o}\,2^{k_o}$ of the staircase are the lowest bits $2^{\nu(Z_{o-1})} = 2^{k_o}$, each times the factors 3 that follow it (by induction on n). The trailing zeros only move up, $\nu(Z_o) \geq \nu(Z_{o-1}) + 1$, and between two odd steps the path is $Z_n/2^j$. So c drops at Order n, the first n at which the odd part of Z_n is below c, that is $Z_n < c\,2^K$ with $K = \nu(Z_n) \geq L$. The margin of the drop is then $c\,(2^K - 3^n)/\rho$ instead of $c\,(2^L - 3^n)/\rho$, and $K - L$ counts the 0-bits that follow the word: those of the drop value.

Measured, on the 31 730 class ids below 2^{20}: 16 045 have $K > L$, and for them the margin grows at least by the factor $1 + 2^L/d_{23}$; for 47, which drops to 46, from between 1 and 2 to above 20. The tight class ids that are left all have $K = L$, an odd drop value: 63 (margin between 1 and 2), 27 (between 3 and 4), 7 (between 4 and 5). Dividing by 2 keeps the numbers small, and so hides that the drop is a race between the trailing zeros, which move up at least one place per odd step, and the leading bits of $3^n c + \rho$.

One extra zero is always enough. Let x have an admissible word of Order $n \geq 2$ and length L, with $n \leq 3x$. Then $\rho < n\,3^{n-1} \leq x\,3^n < x\,2^L$, so $T^L(x) < 2x$ (tStep_iterate_lt_two_mul; Garner's estimate [5], $T^L(x) < 3^n x/2^L + n/3$, gives the same), and when $T^L(x)$ is even, $T^{L+1}(x) < x$ (drop_succ_of_even). Counted by Order, the drop can come after the word only when the value at its end is odd: the hard case of the width identity is the one where no extra zero follows the word.

Key terms introduced in Chapter 4	
bound of a split	$X_u = (2^{L_u} c_v - \rho(u))/3^{n_u}$ for $w = uv$: the class id c of w is at least X_u.
zeros kept	the odd step as $Z \mapsto 3Z + 2^{\nu(Z)}$, the trailing 0-bits kept: $Z_n = 3^n c + \rho$, and c drops at Order n when the odd part of Z_n is below c.

5 The n-th 3-smooth number, without the list

The n-th 3-smooth number $a(n)$ (OEIS A003586 [13]) can be found without listing the $n-1$ before it. The author's PARI program in that entry does it in two stages, with $\log_2 3$ replaced by a fraction P/Q, and Lean proves the program right, in integers, for every $n \geq 1$. It uses only what Chapters 2 and 3 use: the binary lengths $p_2(i)$ of the powers of 3, and the record Orders.

Record fractions

Call P/Q a *record fraction* when Q is an upper record Order with $P = p_2(Q)$ or a lower one with $P = p_2(Q)-1$ (RecordFrac). Such a fraction gives the binary lengths by floors, $\lfloor iP/Q \rfloor + 1 = p_2(i)$ for $1 \leq i < Q$ (div_record_add_one), and the number of powers of 3 below 2^t as $\lfloor tQ/P \rfloor + 1$ for $1 \leq t < P$ (div_record_swap_add_one).

The first stage: the number of binary digits

The 3-smooth numbers with exactly t binary digits are the $2^{t-p_2(i)}\,3^i$, one for each power of 3 below 2^t. So the count $S(k)$ of the 3-smooth numbers below 2^k is k plus a sum of floors $\lfloor tQ/P \rfloor$, a classical floor sum, computed by Euclidean descent (floorSum_eq, smoothCount_eq); this is Mintz's count of paper 2 at the powers of 2. A bisection (bisect_spec) gives the number B of binary digits of $a(n)$, its rank $r = n - S(B-1)$ among the numbers of B digits, and their count $N = S(B) - S(B-1)$.

The second stage: the rank inside the class

Inside the class of B digits, $2^{B-p_2(i)}\,3^i$ for $i < N$, the numbers are sorted by the block factor $f(i) = 3^i/2^{p_2(i)}$: below one of them lie the 3-smooth numbers below 2^{B-1} and those of the class with a smaller block factor (card_smoothBelowAt_class). With u and v the last lower and the last upper record Order below N (a walk from $(1,1)$, recordWalk_spec), the block factor rises along $i = ku \bmod (u+v)$, $k = 0, 1, \ldots$, skipping $i \geq N$ (card_blockFactor_lt_eq, card_mul_mod_lt): the three-gap theorem. The number of kept i among the first k is a difference of two floor sums (classCount_eq), so a second bisection gives the k of rank r, hence i, and $j = B - 1 - \lfloor iP/Q \rfloor = B - p_2(i)$. The program returns (j, i), and exactly $n-1$ of the 3-smooth numbers lie below $2^j 3^i$ (smoothJl_spec, latticeOrd_smoothJl), for every record fraction P/Q with $P > 2h$, $h = 4 \cdot 2^{\lfloor \log_2(n)/2 \rfloor}$. The program computes floor sums of P, Q and Orders only: it forms no 3-smooth number, not even $a(n)$.

For $n = 51$: $S(11) = 47$, so $B = 12$ and $r = 4$. The class of 12 digits has $N = 8$ members, one for each power of 3 below 2^{12}; $u = 7$ and $v = 5$, and the sweep $k \cdot 7 \bmod 12$ gives $0, 7, 2, (9), 4, (11), 6, 1, (8), 3, (10), 5$, the ones in parentheses skipped: 2048, 2187, 2304, 2592, ... in increasing order. The fourth is $i = 4$, $j = 12 - p_2(4) = 5$: $a(51) = 2^5 \cdot 3^4 = 2592$.

Where the fraction comes from

The PARI program takes P/Q from `bestappr`, which returns a convergent. Every semiconvergent p/n of $\log_2 3$ is a record fraction (recordFrac_of_semiconv), so is every convergent (recordFrac_conv), and with one whose numerator exceeds $2h$ the program returns the exponents of $a(n)$ (latticeOrd_smoothJI_conv). That `bestappr` returns convergents of $\log_2 3$, and not of a nearby number, is a matter of PARI's working precision, outside the proof. Without it, the fraction comes from the Stern–Brocot walk towards $\log_2 3$ in integers: a bracket $2^a \le 3^K < 2^b$ for $K = 2^m$, from fixed-point squaring, decides each mediant p/n by $pK \le an$ (below) or $bn \le pK$ (above), and every step it decides is a step of the true walk, so the walk ends at record fractions (sbCertStep_eq, exists_sbCert). For every large m the walk passes any numerator (exists_lt_walkAt), and the program with the least such m returns the exponents of $a(n)$ for every $n \ge 1$ (latticeOrd_smoothJIFind): no convergent is assumed. Shanks [22] computed the continued fraction of a logarithm from the powers alone, and Jackson and Matthews [6] with floor and ceiling in fixed point, where the agreement of the two makes a partial quotient likely, not certain; here every step the bracket decides is proved.

Key terms introduced in Chapter 5	
record fraction	P/Q with Q an upper record Order and $P = p_2(Q)$, or a lower one and $P = p_2(Q) - 1$.
$\boldsymbol{S(k)}$	the number of 3-smooth numbers below 2^k.
A003586	the 3-smooth numbers in the OEIS.

6 The minimal class ids

The smallest class id of each Order is OEIS A122442 [16], with 5 in the place of the class id 1 of Order 1: $a(n)$ is the smallest $x > 1$ whose drop comes at Order $n-1$, so $2, 5, 3, 11, 7, 39, 287, 231, \ldots$ (paper 2, Chapter 3). The entry defines it by the dropping time, and it is the smallest class id of each Order where the class id drops at the end of its own word, which is checked for every term below.

The scan

The terms are found by a *scan*. A residue $x \bmod 2^W$ fixes the first W steps of x; the residues with no admissible prefix within those W steps are kept, 3 524 586 of them for $W = 28$ (measured); for the others the residue fixes the Order. The kept residues are then taken in increasing order of x, round by round, and each x is followed to its drop. As x grows, a term is final as soon as it is found. Each term is checked when it is written: its trajectory reaches its drop exactly at Order $n - 1$, and the drop value is below $a(n)$. The terms $a(1)$ to $a(397)$ are computed so; the last of them is $a(397) = 87\,784\,734\,829\,927$.

Why not a smarter search

The class id of a word is fixed by a congruence modulo 2^L, and that is easy: for one word, $c \equiv -\rho\, 3^{-n} \pmod{2^L}$ (paper 2, Chapter 3). The hard part is the choice of the word. Two attempts to choose it well, and why they fail:

- *Local moves.* Moving the r-th odd step of a word from position k to k', keeping it admissible, changes ρ by $3^{n-r}\, 2^k\, (2^{k'-k} - 1)$, a local change; but it changes the class id by $-3^{-r}\, 2^k\, (2^{k'-k} - 1)$ modulo 2^L: the bits below k stay, all the bits above can change. A descent by such moves is a restart above the first changed position.

- *A solver.* An SMT encoding (M. Winkler's suggestion: Boolean steps, the levels, the Order, the least x by bisection on a bound) reproduces $a(n)$ and its words for Orders 1 to 25 (measured). Its bit-vector solver fixes the low bits of x first, which is the scan by residues again.

Where the problem sits

In terms of the residues, $c \equiv -\sum_o 3^{-o}\, 2^{k_o} \pmod{2^L}$, with the o-th odd step at position k_o: the smallest class id of an Order asks for a choice of positions, under admissibility, that makes a modular sum small. Problems of this shape can be hard because of the bound. Manders and Adleman [10] showed that deciding whether $x^2 \equiv \alpha \pmod{\beta}$ has a solution with $0 \le x \le \gamma$ is NP-complete (their Theorem 2), also when β is given in factored form; without the bound, square roots modulo a factored β are found by the methods of Tonelli and Shanks. Here the congruence for one word is linear and its least solution is the residue itself; what is left is the choice of the word, and no reduction from an NP-complete problem to it is known.

There is also a reason why such a reduction is unlikely. Per Order there is one answer, $a(n)$, so with n written in unary the set of triples $(1^n, i, b)$ with bit i of $a(n)$ equal to b has only polynomially many strings of each length: it is sparse. By Mahaney's theorem [8] (Theorem 3.1 and Corollary 3.6 there), if an NP-complete set is many-one reducible in polynomial time to a sparse set, then P = NP. So computing $a(n)$ is not NP-hard in that sense unless P = NP. Neither statement says how hard $a(n)$ is; both say only that the hardness, if any, is not of the usual kind.

Key terms introduced in Chapter 6	
scan	the odd x taken in increasing order over the residues modulo 2^W with no admissible prefix in W steps, each followed to its drop.
A122442	the smallest class id of each Order, in the OEIS.

7 Three letters: the stop of A324245

Papers 1 and 2 play a game with two letters. OEIS A324245 [17], the map of Vaillant and Delarue [24] (paper 2, Chapter 6), has three *letters*. It acts on n; on the odd number $k = 2n+1$ it reads

$$\texttt{U}\colon k \equiv 3 \pmod 4 \mapsto \tfrac{3k+1}{2}, \qquad \texttt{D}\colon k \equiv 1 \pmod 8 \mapsto \tfrac{3k+1}{4}, \qquad \texttt{S}\colon k \equiv 5 \pmod 8 \mapsto \tfrac{k-1}{4}.$$

The steps `U` and `D` go from k to the next odd value of its path under T. The step `S` does not: it replaces k by the smaller odd number $(k-1)/4$, which has the same next odd value, as $3\,(k-1)/4+1 = (3k+1)/4$. Follow k by these steps and *stop* at the first value below k; let M be the number of steps `U` and `D` up to the stop, the last step included. For $k = 7$ the steps are $7 \to 11 \to 17 \to 13 \to 3$, its *letter word* is `UUDS`, and $M = 3$; for $k = 9$ one step `D` goes to 7, and $M = 1$.

The stop and the drop

Before the stop every odd value reached is at least k, and the odd values of the path of k under T are among them, in order. A stop by `D` is a step of that path: its value is the first odd value of the path below k, and M is the Order of the drop of k. A stop by `S` is not: the path of k under T is still at or above k, and from the stop value on it goes the same way, a *merge* before the drop. So M is at most the Order of the drop, with equality exactly when the stop is by `D` (le_order_of_isUdsStop, isUdsStop_of_S; checked for every odd $k < 2^{22}$). That every odd $k \geq 3$ has a stop is equivalent to every path under T reaching 1 (stops_iff): a stop by `D` lands on a smaller value of the path, a stop by `S` on a smaller value with the same next odd value, so an induction over the values goes through. So the early stop is equivalent to the Collatz conjecture, and comes sooner: at a merge, before the drop. The number 7 drops at Order 4, to 5, one odd step after its merge with the path of 3.

The words

Write `U` as `1`, `D` as `10` and `S` as `00`. The letters multiply k by about 3/2, 3/4 and 1/4: the factors of the bits of a word of paper 2. After a prefix of the letters the value is $P\,k + c$, with P the product of the factors and c fixed by the prefix, so for all large k of a class the stop comes at the first prefix with $P < 1$ (isUdsStop_iff_of_large). It does so for every odd $k < 2^{22}$ (measured); whether it does for every k is open. At the last letter of a class that stops by `D` it follows from the width identity of the word at which the class drops (ClassDrops; paper 2, Chapter 7). A prefix with $P < 1$ stops every $k > 2^e 3^m$ of its class, e its bits and m its letters `U` and `D` (stop_of_coeff), and the odd k modulo 2^{E+1} with no such prefix of at most E bits are a part of the classes that goes to 0 as E grows (density; Terras [23], Everett [4]).

Call a letter word one *of the product rule* when it ends at the first prefix with $P < 1$. By the product rule its bits are an admissible word of Order M, or one followed by `0`. A `1` followed by z zeros splits in exactly one way into letters: `U` and $z/2$ letters `S` when z is even, `D` and $(z-1)/2$ letters `S` when z is odd. The first product below 1 falls at the end of an admissible word or one bit later, and both occur for every admissible word. So for $m \geq 1$ there are $2\,\mathrm{A100982}(m)$ [15] letter words of the product rule with $M = m$ (paper 2, Chapter 5), $\mathrm{A100982}(m)$ of them fixing k modulo $2^{p_2(m)+1}$ and as many modulo $2^{p_2(m)+2}$ (card_udsStopSet, udsWord_class; counted for $m \leq 16$). For $k = 7$ the bits are `111000`, the admissible word `11100` of Order 3 followed by `0`. These bits need not be the word of k: in the letters, the steps `S` of an odd value come before its `U` or `D`.

The least odd k with $M = m$, for $m = 1, \ldots, 15$: 3, 11, 7, 39, 95, 155, 191, 127, 603, 359, 239, 159, 639, 283, 743 (measured, $k < 2^{22}$). For $m = 1$ to 4, 7, 8 and 11 to 14 this is the least x that drops at Order $m + 1$ (Chapter 6).

Key terms introduced in Chapter 7	
U, D, S	the three steps on odd k: $(3k+1)/2$ for $k \equiv 3 \pmod 4$, $(3k+1)/4$ for $k \equiv 1 \pmod 8$, $(k-1)/4$ for $k \equiv 5 \pmod 8$.
stop	the first value below k reached by the letters.
letter word	the letters of k up to its stop; in bits $\boxed{1}$, $\boxed{1\,0}$, $\boxed{0\,0}$.
M	the number of letters `U` and `D` up to the stop.
merge	a stop by `S`: the next odd value of the stop value lies on the path of k under T, before its drop.
A100982	the number of admissible words of each Order, in the OEIS.
A324245	the map of the three letters, on $n = (k-1)/2$, in the OEIS.

8 The ruler gap

This chapter is an open statement on neighbouring 3-smooth numbers, with its reductions. Papers 1 and 2 do not use it, and nothing in this paper rests on it.

The statement

Draw the 3-smooth numbers on a *plain ruler*: v at distance v from the left, in the row of its number of factors 3. Let $a < b$ be neighbours among them. A second step with the same factor would land at b^2/a; the straight line from a through b, continued over the same number of rows, arrives at $2b-a$, which is before it (line_falls_short). The *ruler gap* says that no 3-smooth number lies strictly between $2b - a$ and b^2/a (RulerGap). For $a = 8$, $b = 9$ the gap is from 10 to $81/8$. Under the ruler gap a 3-smooth number lies before the second step exactly when it lies at or before the end $2b - a$ of the line (rule_iff_of_rulerGap). The same rule with the place where the continued line meets row 0 is false: from 3^{11} the step to $2^8 \cdot 3^6$ is to a neighbour, its line meets row 0 at $197\,996 + 2/5$, and the second step of that factor lands at $2^{16} \cdot 3$, before that place (border_rule_fails).

In exponents and record Orders

The statement is one on exponents only (rulerGap_iff_ordersGap). A step between neighbours that raises the exponent of 2 is a step of an upper record Order b, by the factor $2^{p_2(b)}/3^b$; one that raises the exponent of 3 is a step of a lower record Order a, by $3^a/2^{p_2(a)-1}$ (stepAt_two_three_iff, stepAt_three_two_iff; these are the two steps of Chapter 2), and the ruler gap is a statement on those steps (rulerGap_iff_records). For a step of an upper record Order b, a number in its gap is one more factor $3^v/2^u$ with $u+1 = p_2(v)$, where b and v split at $K = b+v$, $p_2(K)+1 = p_2(b)+p_2(v)$, and

$$1 - f(K) < \big(1 - f(b)\big)^2$$

for the block factor f (gapAt_two_three_iff_deficit); the lower record Orders have their own inequality of this kind. So the ruler gap is the failure of these deficit inequalities wherever the step is due (rulerGap_iff_deficit), and the windows where a step is due end at the next record Order. Over each window the block factor is extreme at one record Order c, so the ruler gap is one inequality per pair of neighbouring record Orders $b < b'$, at the largest upper record Order $c \le b + b'$, and likewise for the lower ones (rulerGap_iff_pairs). In words: the deficit $1 - f$ at such a c is not below the square of the deficit at b. That is a statement on how closely the fractions on the Stern–Brocot walk approach $\log_2 3$, and it is open.

The finite check

Lean's kernel checks the pair inequalities for every pair of neighbouring record Orders up to 10^{60} (upperPairs_upTo, lowerPairs_upTo), with the certified walk of Chapter 3, here at $K = 2^{500}$ (rulerUpWalk_ok, rulerLowWalk_ok). With an upper and a lower record Order between 10^{59} and 10^{60} (exists_records_between), the ruler gap holds for every exponent of 2, with every exponent of 3 up to 10^{59} (rulerGapUpTo_ten_pow). This is a finite check; the ruler gap itself stays open.

Key terms introduced in Chapter 8	
plain ruler	v drawn at distance v, in the row of its number of factors 3.
ruler gap	for neighbouring 3-smooth numbers $a < b$: none lies strictly between $2b - a$ and b^2/a; open (RulerGap).

9 What is left out

The map $3x - 1$

On the negative numbers T is the map $x \mapsto (3x - 1)/2$ for odd x, $x/2$ for even x, with the sign turned. There the drop has a plain answer: $5 \to 7 \to 10 \to 5$ never falls below 5. Compared once with T word by word, it showed the change of sign only, no link between the classes (measured). It says nothing on the drop under T, and this paper leaves it out.

Shares and ratios

Shares, such as how many of the starts of Chapter 7 stop by `D`, and ratios seen on a range are not statements that hold for every number. They are left out.

Other multipliers

Other odd multipliers $q \geq 3$ are left out: paper 2 proves its chain decomposition and sweeps for every odd q, the setting of M. Winkler's admissible sequences [25], and the lean-files TBracketCountQ, TChainParseQ and TBracketDropQ carry the rest of paper 2's Chapter 4 over to every odd q.

A The Lean formalization

The Lean names this paper cites are proved in the Lean 4 project AcceleratedCollatz of papers 1 and 2 (paper 2, Appendix A), without `sorry`; open statements and theorems taken from the literature enter as named hypotheses (ClassDrops, RulerGap, RhinBound, RhinInt), and the range check of Chapter 3 as a hypothesis of the theorems that use it. Table 2 lists the lean-files by chapter; the names of the PowDiff files are in the namespace PowDiff, those of the two A349414 files, self-contained extracts, in namespaces of their file names. Chapter 6 is computed, not formalized.

Ch.	lean-file	content
1, 2	TRho{Bound,Lower}, TSmooth{Between,Next,Place,Succ}, TStaircase, TWidth, TMonksQ	staircases, the bounds on ρ, the 3-smooth numbers in order and between 3^n and $2^{p_2(n)}$, the width identity, ρ over the power series in q
3	TChristoffel, TRhin{Bound,Check}, TRho{Max,Slack}, TRhoMax{Bound,Check,Record}, TFiniteCheckNext	the largest ρ, the first word as a Christoffel word, its slack, the finite reduction and its records, the Rhin check, the width identity up to Order 64 497 106 given the range check
3	PowDiff{,Far,Rhin,Walk}, PowDiffFar{Bracket,Check,Nodes}, PowDiffFar{Low,Rhin}{1..4}	the lower side to 10^{60} and both sides to 10^{200}, the differences $\lvert 2^i - 3^j \rvert$ (A014121, A363999)
4	TCongruenceSplit, TPrefixBlockFail, TGap{Ratio,Record}, TUpperRecord, TA398772, TEvenDrop, TRhinSplit, TGersonides	ρ of a concatenation, the bound of a split, the failure of the block statement, the gap ratio and its records, the sieve of the record search, the upper record Orders, the even case, $3c+1 \le 2^L$ under Rhin's bound, the powers of 2 and 3 that differ by 1
5	TBisect, TFloorSum, TRecord{Floor,Walk}, TSmooth{Program,ProgramConv,Rank}, TSternBrocotCert	the A003586 program
7	TUds{,Count,Drop}, A349414{Count,Run}	the three letters: the stop and the drop, the stop for large k, the classes, the count; the stop for every k and the Collatz conjecture, the density
8	SmoothRuler, TRuler{Check,Deficit,Pair,Record}	the ruler gap, its reductions and the finite check
9	TBracketCountQ, TChainParseQ, TBracketDropQ	every odd q: the bracket words, the parse, the brackets and the map $qx+1$

Table 2: The lean-files of this paper, by chapter; braces group names as in a shell, so TRuler{Check,Pair} is TRulerCheck and TRulerPair.

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